

*MA.912.AR.1.7***Benchmark****MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.**

Example: The expression $4x^3y - 3x^2y^4$ is equivalent to the factored form $x^2y(4x - 3y^3)$.

Example: The expression $16x^2 - 9y^2$ is equivalent to the factored form $(4x - 3y)(4x + 3y)$.

Benchmark Clarifications:

Clarification 1: Within the Algebra I course, polynomial expressions are limited to 4 or fewer terms with integer coefficients.

Connecting Benchmarks/Horizontal Alignment

- MA.912.NSO.1.1
- MA.912.NSO.1.2
- MA.912.AR.3.1, MA.912.AR.3.5, MA.912.AR.3.6, MA.912.AR.3.7, MA.912.AR.3.8

Terms from the K-12 Glossary

- Expression
- Polynomial
- Greatest Common Factor

Vertical Alignment**Previous Benchmarks**

- MA.6.AR.1.4
- MA.7.AR.1.2
- MA.8.AR.1.3

Next Benchmarks

- MA.912.AR.1.8

Purpose and Instructional Strategies

In grade 8, students rewrote binomial algebraic expressions as a common factor times a binomial. In Algebra I, students rewrite polynomials, up to 4 terms, as a product of polynomials over the real numbers. In later grades, students will rewrite polynomials as a product of polynomials over real and complex number systems.

- Instruction includes special cases such as difference of squares and perfect square trinomials.
- Instruction builds upon student prior knowledge of factors, including greatest common factors.
- Instruction includes the student understanding that factoring is the inverse of multiplying polynomial expressions.

- Instruction includes the use of models, manipulatives and recognizing patterns when factoring.
 - Sum-Product Pattern
The expression $x^2 + 7x + 10$ can be written as $(x + 5)(x + 2)$ since $5 + 2 = 7$ and $5(2) = 10$.
 - Factor by Grouping
The expression $x^3 + 7x^2 + 2x + 14$ can be grouped into two binomials and rewritten as $(x^3 + 7x^2) + (2x + 14)$. Each binomial can be factored and rewritten as $x^2(x + 7) + 2(x + 7)$ resulting in the same factor and the factored form as $(x^2 + 2)(x + 7)$.
 - A-C Method
When factoring trinomials $ax^2 + bx + c$, multiply a and c , then determine factor pairs of the product. Using the factor pair that adds to b and multiplies to c , rewrite the middle term and then factor by grouping.
 - For example, given $2x^2 + x - 6$ and that $ac = -12$, one can determine that two numbers that add to 1 and multiply to -12 are 4 and -3 . This information can be used to rewrite the given quadratic as $2x^2 + 4x - 3x - 6$. Then, using factor by grouping the expression is equivalent to $(2x^2 + 4x) - (3x + 6)$ which is equivalent to $2x(x + 2) - 3(x + 2)$ which is equivalent to the factored form $(2x - 3)(x + 2)$.
 - Area Model/Box Method
To factor $ax^2 + bx + c$, the general box method is shown below.

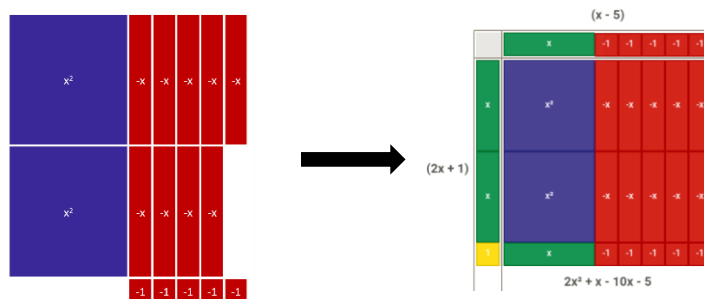
	ax^2	b_1x
b_{2x}		c

- For example, to factor $2x^2 - 9x - 5$ the box method is shown below. The first term, $2x^2$, is placed in the top left box and the constant, -5 , is placed in the bottom right box. Then reference the A-C Method or guess and check to determine how to separate the middle term, $-9x$ into the remaining boxes. The factors can then be identified using the Greatest Common Factor across each row and column.

	$2x$	1
x	$2x^2$	$1x$
-5	$-10x$	-5

○ Area Model/Box Method (Algebra tiles)

The factorization of $2x^2 - 9x - 5$ using algebra tiles is shown below. Begin with the representation of the given polynomial and arrange the tiles into a complete rectangle, adding zero pairs of tiles if needed. The factors are then determined by the dimensions of the rectangle.



Common Misconceptions or Errors

- Students may not identify the greatest common factor or factor completely.

Strategies to Support Tiered Instruction

- Instruction includes providing a flow chart to reference while completing examples.
- Instruction includes providing definition of greatest common factor and strategies for identifying the greatest common factor of numerical or algebraic terms.
 - For example, the expression $8x^3 - 4x^2$ has common factors of 2 and x , but these are not greatest common factors. The greatest common factor of the coefficients is 4 and the greatest common factor of the variable terms is x^2 . So, the greatest common factor of the two terms is $4x^2$. The expression $8x^3 - 4x^2$ can be rewritten as $4x^2(2x - 1)$.

Instructional Tasks

Instructional Task 1 (MTR.3.1, MTR.4.1, MTR.5.1)

Part A. Given the polynomial $x^4 - 16y^4z^8$, rewrite it as a product of polynomials.

Part B. Discuss with your partner the strategy used. How do your polynomial factors compare to one another?

Instructional Task 2 (MTR.3.1, MTR.5.1)

Part A. What are the factors of the quadratic $16x^2 - 48x + 36$?

Part B. Determine the roots of the quadratic function $f(x) = 16x^2 - 48x + 36$.

Part C. What do you notice about your answers from Part A and Part B?

Part D. Graph the function $f(x) = 16x^2 - 48x + 36$.

Instructional Items

Instructional Item 1

Given the polynomial $x^4 - 16y^4z^8$, rewrite it as a product of polynomials.

Instructional Item 2

Given the polynomial $x^2 - 10x + 24$, rewrite it as a product of polynomials.

Instructional Item 3

Given the polynomial $x^3 - 3x^2 - 9x + 27$, rewrite it as a product of polynomials.

Instructional Item 4

What is one of the factors of the polynomial $21r^3s^2 - 14r^2s + 14rs^3$?

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*